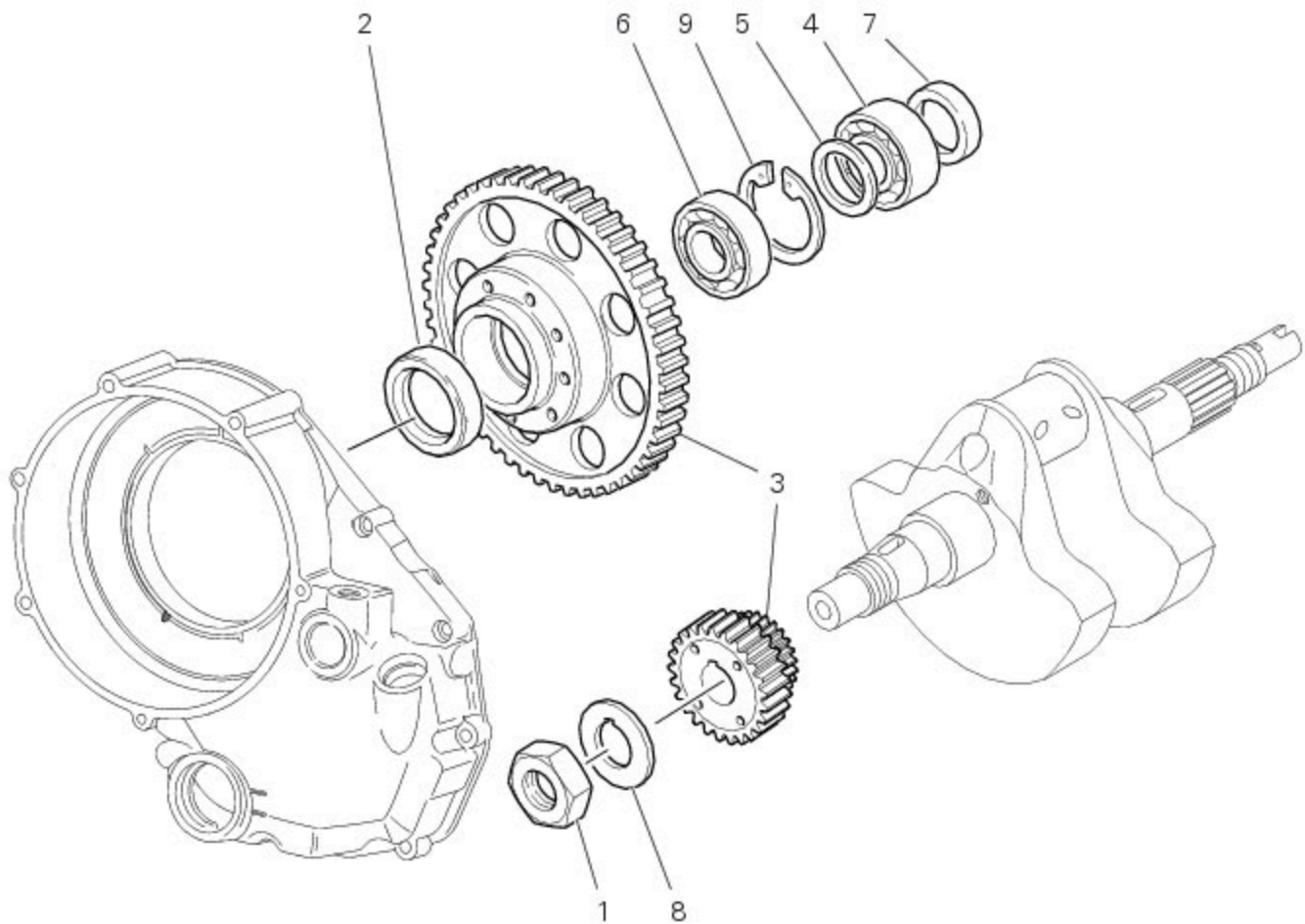


6.3 -Clutch assembly: primary drive gears



- 1 Nut
- 2 Sealing ring
- 3 Primary drive gears
- 4 Bearing
- 5 Circlip
- 6 Bearing
- 7 Spacer
- 8 Lock washer
- 9 Circlip

 Spare parts catalogue

1100 ABS [CLUTCH](#)
1100 ABS [CONNECTING RODS](#)
1100 S ABS [CLUTCH](#)
1100 S ABS [CONNECTING RODS](#)

Important

Bold reference numbers in this section identify parts not shown in the figures alongside the text, but which can be found in the exploded view diagram.

Removal of the primary drive gear

Operation	Section reference
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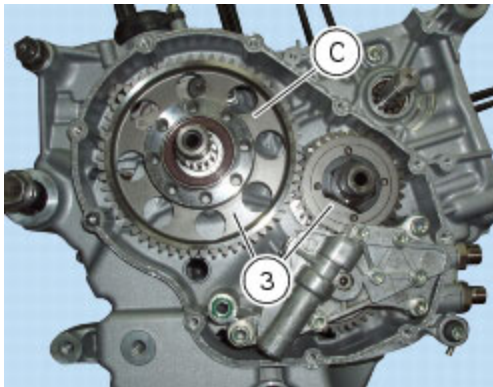
Drain the engine oil	D 4, Changing the engine oil and filter cartridge
Remove the clutch assembly	N 6.1, Removal of the clutch
Remove the clutch cover	N 6.2, Removal of the clutch cover



Note

For clarity, the figures show the engine removed from the frame.

Remove the driven gear (C) of the primary drive (3) complete with the bearings and oil sealing ring.



To renew the inner parts of the gear, use a suitable drift (A) and a support (B).

After removing the oil sealing ring (2), tap from the inside towards the exterior using part of the inner race of bearing (4) to be removed, after moving aside spacer (5) located between the two bearings.

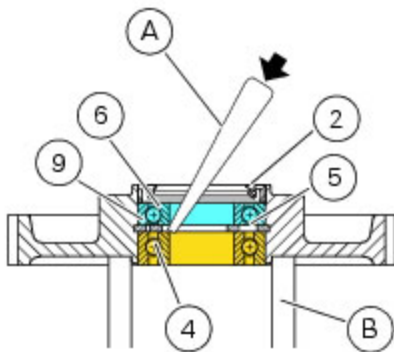
Tap on different points to remove the components squarely.

Use the same technique to remove the bearing (6).



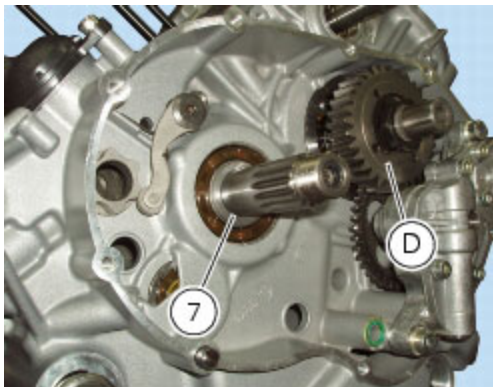
Important

Once disturbed, the oil sealing ring (2), circlip (9) and the spacer (5) must be renewed. The latter two components must always be renewed as a pair.

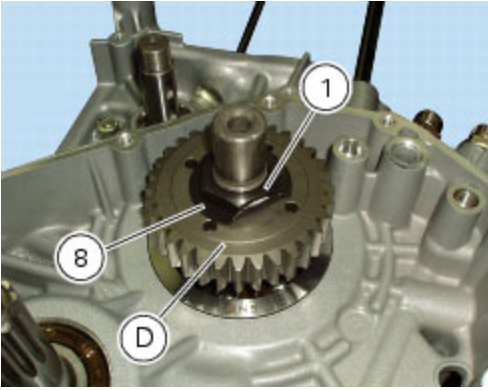


Remove the spacer (7) from the gearbox primary shaft.

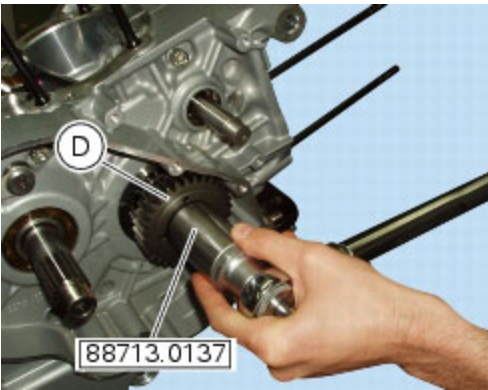
Remove the oil pump (Sect. N 2.1, [Removal of the oil pump](#)) in order to be able to disengage the driving gear (D).



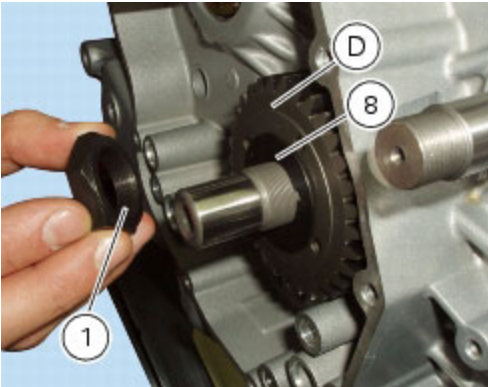
Straighten out the lock washer (8) on the nut (1) securing the driving gear (D) of the primary drive pair.



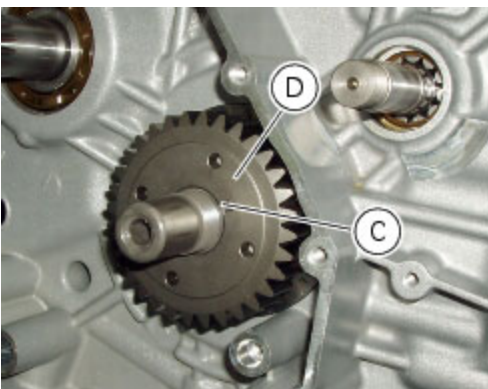
Install the tool **88713.0137** on the gear (D) and hold it against rotation by inserting a pin in one of the engine mounting bolt holes.



Loosen the retaining nut (1) of the driving gear (D) using a socket wrench of suitable length. Remove the nut (1) and the lock washer (8).

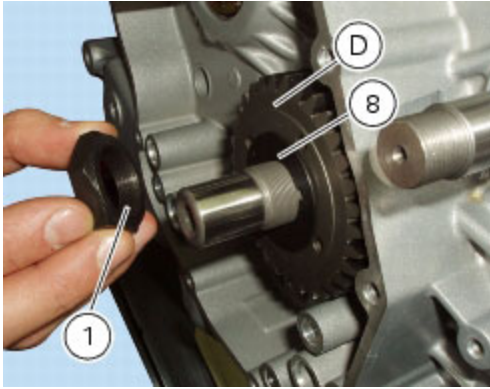


Remove the primary driving gear (D) using puller **88713.2092** and interposing a packing piece of aluminium or copper between the crankshaft and the screw of the puller. Take care not to damage the tongue (C) located on the crankshaft.

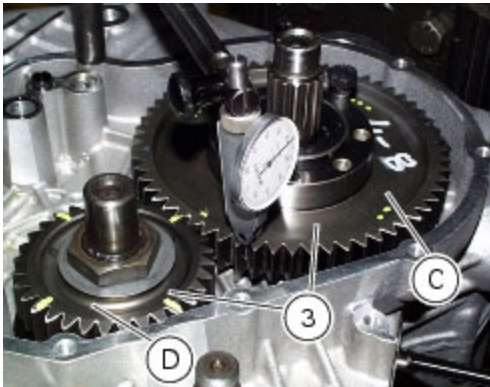


Fitting the primary drive gear pair and checking backlash

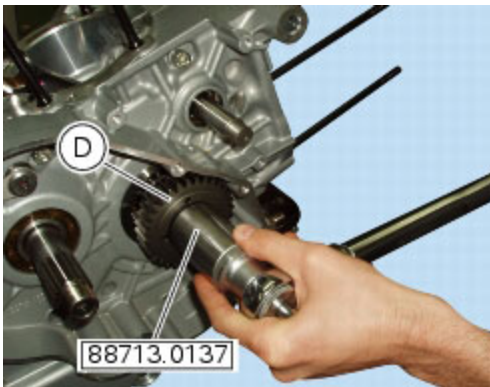
Carefully degrease the splined end of the crankshaft and the corresponding internal splines of the primary gear.
Fit the gear (D) onto the crankshaft with oil pump drive sprocket facing the crankcase.
Temporarily secure the gear with the washer (8) and nut (1).



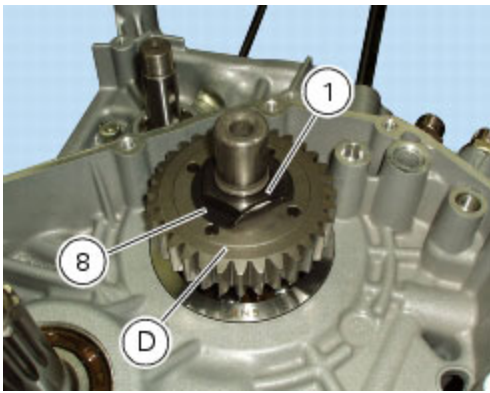
If fitting a new primary drive gear pair (3), check the backlash.
Temporarily fit the gear (C) complete with the bearings and the oil sealing ring onto the gearbox primary shaft and fix a dial gauge on the crankcase; position the dial gauge stylus on a gear tooth.
Turn the driven gear (D) to mesh the teeth and check with the dial tool that backlash ranges between **0.05** and **0.07** mm.
Measure the backlash at 16 diametrically opposed points of the gear.
If the measured values are outside the permissible tolerance, change the position of the driven gear (C) on the primary shaft leaving crankcase gear (D) untouched. If still outside tolerance values, renew the primary drive gear pair.



When this test is complete, finally tighten the previously installed nut (1).
Install the tool **88713.0137** on the gear (D) and hold it against rotation by inserting a pin in one of the engine mounting bolt holes.
Tighten the nut to the specified torque with a torque wrench (Sect. C 3, [Engine torque settings](#)), turning clockwise.



Stake the washer (8) on the gear (D) in correspondence with the milled location and on the nut (1) in a diagonally opposed position.



It is good practice to fill the crankshaft with oil so that it does not operate dry during its first revolutions. Refit the oil pump and check the clearance between the oil pump gear and primary drive gear on the crankshaft (Sect. N 2.1, [Refitting the oil pump](#)).

Operation	Section reference
Refit the clutch cover	N 6.2, Refitting the clutch cover
Refit the complete clutch assembly	N 6.1, Reassembling the clutch
Refill the engine with oil	D 4, Changing the engine oil and filter cartridge